Practice Exam - 1

Question 64 of 117

A 69-year-old woman (body mass index 29, blood type O positive) with long-standing nonischemic cardiomyopathy and type II diabetes mellitus (HbA1c 6.5%), is evaluated for worsening fatigue, dyspnea at rest, and orthopnea over the past 2 months. She has had three hospitalizations for heart failure in the last six months, including the most recent 1 month prior. Currently, she develops dyspnea walking to her mailbox. She is frustrated about her inability to participate in family activities. Approximately a year ago, she had been walking 2-3 miles daily with her dog. Her cardiac medications include carvedilol 3.125 mg twice daily, dapagliflozin 10 mg daily, spironolactone 25 mg daily, sacubitril-valsartan 24-26 mg twice daily, and bumetanide 4 mg once daily. She was previously tolerating sacubitril-valsartan 97-103 mg twice daily, but the dose was lowered due to symptomatic hypotension. Her vital signs today are BP 92/44 mmHg HR 84 bpm RR 24 breaths per min SpO2 93% at room air. Her exam shows JVP 10 cmH2O, normal S1, S2, no murmurs, rubs or gallops, clear lungs to auscultation bilaterally, and 1+ ankle edema bilaterally. 12-lead EKG shows RBBB with QRS duration of 134 ms. Echocardiogram shows severe biventricular dysfunction, LVEF 15%, LVESD 75 mm, moderate to severe mitral regurgitation, and estimated pulmonary systolic pressure 55 mmHg. Cardiopulmonary stress testing indicates that she has peak oxygen consumption of 10.9 mL/kg/min (46% predicted) at a respiratory exchange ratio of 1.10.

A. Refer for cardiac resynchronization therapy and implantable cardioverter defibrillator placement (CRT-D)

B. Refer for transcatheter edge-to-edge mitral valve repair (TEER)

C. Refer the patient for palliative care consultation

D. Refer the patient to an advanced heart failure center for evaluation of advanced therapies

Commentary | References

Testing Point: The board-certified AHFTC specialist should recognize when a patient with Stage D heart failure requires prompt referral for evaluation of advanced heart failure therapies.

Correct Answer: D. Refer the patient to an advanced heart failure center for evaluation of advanced therapies

Rationale: This patient has ACC/AHA Stage D heart failure and endorses NYHA Class IV symptoms. Given her symptom progression, frequent hospitalizations and intolerance of guideline directed medical therapy, she should be evaluated for advanced therapies, including heart transplant.

The other answer choices are incorrect.

Answer A: She is not a suitable candidate for CRT-D placement at this time given that she has non-LBBB, QRS < 150 msec, and less than 1 year survival based on her CPST results.

Answer B: Similarly, she is not a suitable candidate for transcatheter mitral repair as she does not meet criteria used in the COAPT trial, which included symptomatic patients with moderate to severe secondary mitral regurgitation with LVEF 20 to 50%, LVESD < 70 mm, and PASP < 70 mmHg. Her LVEF is <20% with excessive LV dilatation, making her unlikely to benefit from TEER at this time.

Finally, as long as it is consistent with her goals of care, she merits an evaluation for heart transplant. There are no obvious contraindications based on the stem. Her RV dysfunction may preclude durable LVAD placement at some centers.

Answer C: A palliative care consult should be part of the evaluation for advanced therapies, but focusing on this exclusively may delay the time sensitive window of potential eligibility for advanced therapies. As she clearly wishes to be able to participate in her family's activities, proceeding with evaluation for heart transplant at this time is the best option.